

LEARMONTH, Australia

WMO: 943020

Lat: 22.2406S

Lon: 114.0967E

Elev: 18

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	48.0	50.1	21.1	15.8	80.3	25.9	19.9	78.0	22.8	69.3	20.8	69.7	7.0	230	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.1	106.2	69.7	103.3	69.8	100.5	70.1	79.6	87.7	78.6	87.2	77.7	86.9	13.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.6	143.5	83.1	76.6	138.9	82.7	75.2	132.4	82.6	43.3	87.7	42.2	87.1	41.2	86.7	91.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.4	21.6	20.1	DB	42.6	113.3	1.9	2.6	41.2	115.2	40.1	116.7	39.0	118.2	37.6	120.0	(4)
(5)			WB	38.2	81.7	1.9	2.5	36.9	83.4	35.8	84.9	34.7	86.3	33.4	88.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.4	86.3	86.6	85.3	80.1	72.0	65.8	64.1	66.5	70.6	76.5	79.7	83.7	(6)	
	DBStd	8.99	4.22	3.99	3.97	4.00	4.34	4.11	3.60	3.69	3.70	4.07	4.29	4.59	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	128	0	0	0	0	2	39	60	25	2	0	0	0	(9)	
	CDD50	9628	1124	1025	1094	904	682	474	437	512	618	821	891	1046	(10)	
	CDD65	4281	659	605	629	454	219	63	32	72	170	356	441	581	(11)	
	CDH74	48514	8129	7645	7802	4901	1855	350	244	752	1717	3675	4696	6748	(12)	
(13)	CDH80	25404	4748	4320	4297	2356	552	23	19	137	579	1854	2565	3954	(13)	
(14)	Wind	WSAvg	10.9	13.6	12.3	11.2	9.3	8.1	8.1	9.0	10.9	12.7	13.3	13.7	(14)	
(15) Precipitation	PrecAvg	9.1	1.3	1.5	1.3	0.7	1.6	1.8	0.8	0.5	0.1	0.1	0.0	0.0	(15)	
	PrecMax	24.2	14.2	9.4	12.0	5.1	9.1	8.8	3.9	2.5	0.6	1.8	1.3	0.4	(16)	
	PrecMin	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	5.3	2.8	2.3	2.4	1.1	1.9	1.8	1.0	0.7	0.2	0.3	0.2	0.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	111.3	109.3	106.2	101.3	93.3	82.6	82.4	87.8	95.0	101.1	104.2	109.6	(19)	
		MCWB	69.1	71.0	69.9	67.3	63.3	63.0	59.8	59.0	61.9	64.3	66.1	69.4	(20)	
	2%	DB	105.9	104.7	102.4	97.1	88.8	79.9	79.0	83.8	89.6	96.7	99.9	104.3	(21)	
		MCWB	70.2	71.5	70.6	66.7	63.9	62.7	60.3	59.3	60.7	64.2	66.1	67.9	(22)	
	5%	DB	102.1	101.4	99.4	94.0	85.6	77.8	76.8	80.7	86.0	93.4	96.6	100.5	(23)	
		MCWB	71.2	72.3	71.2	67.3	64.5	62.4	59.7	59.6	60.9	63.5	65.9	68.6	(24)	
	10%	DB	98.3	97.8	96.3	91.2	82.7	75.8	74.6	78.4	82.8	89.9	93.0	96.9	(25)	
		MCWB	72.1	73.1	71.7	68.1	65.2	62.2	59.6	59.3	60.7	63.1	66.0	69.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.8	80.9	80.7	78.1	75.4	72.7	70.6	69.5	69.7	72.1	74.3	78.9	(27)	
		MCDB	88.3	89.1	87.6	83.6	79.7	76.1	74.5	75.7	79.5	83.8	86.7	87.0	(28)	
	2%	WB	79.1	79.6	79.1	76.3	73.3	70.5	67.9	67.1	67.5	69.7	72.4	76.5	(29)	
		MCDB	87.7	88.1	87.1	82.9	78.5	74.1	71.7	73.7	76.9	82.1	85.4	88.6	(30)	
	5%	WB	78.0	78.7	78.0	74.6	71.4	68.4	65.9	65.0	66.0	68.0	70.8	74.7	(31)	
		MCDB	87.7	87.6	86.4	82.4	77.8	72.7	70.5	73.0	76.3	81.6	84.3	88.3	(32)	
	10%	WB	76.8	77.7	76.8	73.0	69.5	66.4	63.6	63.0	64.3	66.6	69.4	73.0	(33)	
		MCDB	87.9	87.2	85.7	82.3	76.5	72.0	69.5	72.2	76.1	80.4	82.8	87.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	24.6	22.1	22.3	22.1	21.7	20.5	22.0	25.1	26.7	28.3	27.0	26.7	(35)	
		MCDBR	31.4	28.8	27.0	26.3	25.3	22.1	24.7	27.7	30.5	32.7	32.0	32.4	(36)	
		MCWBR	7.2	7.2	7.4	9.0	11.1	10.5	11.0	11.6	10.2	8.2	7.6	7.3	(37)	
	5% WB	MCDBR	19.5	17.9	17.8	18.8	17.5	14.2	16.8	22.8	25.0	27.5	26.0	23.9	(38)	
		MCWBR	7.1	6.7	7.8	9.7	10.5	8.9	10.5	12.5	11.2	9.6	8.6	8.1	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.386	0.389	0.370	0.349	0.325	0.318	0.312	0.311	0.323	0.342	0.346	0.360	(40)	
	taud		2.521	2.529	2.575	2.598	2.616	2.619	2.616	2.593	2.538	2.504	2.518	2.541	(41)	
	Ebn at Noon		304	299	297	290	285	281	287	298	306	310	314	312	(42)	
	Edh at Noon		36	35	32	29	26	25	26	29	33	36	36	35	(43)	
(44) All-Sky Solar Radiation	RadAvg		2506	2372	2087	1734	1443	1269	1355	1705	2084	2391	2600	2634	(44)	
	RadStd		107	118	124	99	90	78	113	65	60	67	62	114	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.56	N/A	-1.84	N/A	N/A	N/A	N/A	N/A	+275	+288
Regional (3 neighbors)	+0.56	N/A	-1.84	N/A	N/A	N/A	N/A	N/A	+275	+288

Nomenclature: See separate page